

Yoyo's Club - Synthesis Report

What is this project

Yoyo is a nightclub owner who manages everything manually — staff schedules on paper, salary calculations by hand, and attendance tracked in a notebook. The goal was to build a proper web dashboard so he can manage everything from one place — staff, payroll, tickets, attendance, and employee clock-in all in one system.

What I did

I was assigned the DevWeb module so I build the full admin panel. Here is what I made:

- Login page with email and password, JWT token authentication, role-based redirect — admin goes to dashboard, employee goes to their own page
- Dashboard with 4 stat cards (total staff, DJs, bouncers, estimated payroll), attendance chart with Today/Week/Month/Year filters and custom date range picker, navigation arrows to scroll through chart history, staff overview chart with hours and payroll toggle and role filter
- Staff page where Yoyo can hire employees (creates employee record and login account at the same time), edit details, fire employees, see real clock-in hours and real pay per employee, live session timer counting up for each currently clocked-in employee, force clock out remotely, reset only today's hours for fraud prevention, pay via Stripe with duplicate payment warning
- Payroll page with full payment history, total paid all time, paid this week stats, auto-resets employee clock hours to zero after payment
- Tickets page to create events with quick day selector, sell entry tickets, track sold vs remaining, revenue per event, attendance chart auto-updates when tickets are sold

And also full Employee panel. Here is what I made:

- Separate employee dashboard accessible only to employees
- Clock in and clock out with live status indicator
- View own details — role, contract, hourly rate, real hours worked, estimated pay
- Next payment date (always next Monday)
- Full clock-in history table
- Full payment history table

Tech stack

- Frontend: React + Vite, Recharts, Axios
- Backend: FastAPI with Python
- Database: PostgreSQL on Supabase (cloud)
- Auth: JWT token + sha256 password hashing
- Payment: Stripe sandbox

How salary calculation works

I made 3 different formulas based on contract type:

- Full-time: hours x rate + 10 percent bonus
- Part-time: hours x rate (no change)
- Freelance: hours x rate minus 20 percent tax

Database tables

Table	What it stores
employees	Role, contract, hourly rate, hours worked
users	Email, hashed password, role (admin or employee)
attendance	Guest count per date
tickets	Event name, date, price, total and sold tickets
payments	Payment history records per employee
clockins	Clock-in and clock-out per employee per day

Problems I faced

- Supabase direct URL was not resolving on my network so I switched to pooler URL and it worked
- Python 3.14 has issue with bcrypt so I switched to sha256_crypt
- I had duplicate route in tickets.py so Python was using wrong version, I fixed by rewriting the file clean
- Attendance was adding old data to new so I added reset logic when new event is created

Design pattern

I followed MVC pattern throughout the project:

- **Model** — SQLAlchemy table classes in **models.py** define the database structure
- **Controller** — FastAPI route functions in **routes/** folder handle all API logic
- **View** — React pages in **src/pages/** folder handle the UI